

REINFORCE++: STABILIZING CRITIC-FREE POLICY OPTIMIZATION WITH GLOBAL NORMALIZATION

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ABSTRACT

Reinforcement Learning from Human Feedback (RLHF) plays a crucial role in aligning Large Language Models (LLMs). The dominant algorithm, Proximal Policy Optimization (PPO), employs a critic network to estimate advantages, which introduces significant computational and memory overhead. To address this, a family of critic-free algorithms (e.g., GRPO, RLOO) has emerged. However, methods such as GRPO rely on *prompt-level (local)* advantage normalization, whose finite-sample distortion we characterize and whose training stability and held-out generalization we examine. To solve these challenges, we introduce **REINFORCE++**, a critic-free framework centered on **Global Advantage Normalization**. By normalizing advantages across the entire global batch rather than small, prompt-specific groups, our method uses a shared scale whose sample moments converge under independent sampling; this does not imply unbiasedness of the full policy gradient. We introduce two variants: **REINFORCE++**, a computationally efficient algorithm ($k \geq 1$) for general-domain RLHF, and **REINFORCE++_{w/ Baseline}**, a group-sampling variant ($k > 1$) for complex reasoning tasks. Across our evaluated settings, each variant improves training stability and matches or exceeds strong critic-free baselines; in complex agentic settings, **REINFORCE++_{w/ Baseline}** is competitive with PPO.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) is a key technique for aligning Large Language Models (LLMs) with human values and preferences (Vemprala et al., 2023; Achiam et al., 2023; Ouyang et al., 2022; Shen & Zhang, 2024; Shen et al., 2025; Hu et al., 2024). Despite the emergence of non-RL alternatives like DPO (Rafailov et al., 2023), state-of-the-art applications such as ChatGPT/GPT-4 (Vemprala et al., 2023; OpenAI, 2023), Claude (Anthropic, 2023), and Gemini (Team et al., 2023) continue to rely on RL algorithms.

The dominant RL algorithm in this space is Proximal Policy Optimization (PPO) (Schulman et al., 2017). PPO employs an "Actor-Critic" architecture, where a dedicated critic network is trained to estimate the advantage function (i.e., the extent to which an action yields a higher expected return than the mean). However, this critic network introduces substantial computational overhead and memory requirements, making training prohibitively expensive and limiting large-scale model alignment on small-scale clusters (Shao et al., 2024).

To address PPO’s efficiency issues, a family of REINFORCE-based methods has emerged without the critic model, including ReMax (Li et al., 2023), RLOO (Ahmadian et al., 2024), and GRPO (Shao et al., 2024). These algorithms replace the critic network with an estimator that computes the advantage from multiple responses to the same prompt.

However, this critic-free approach introduces a critical challenge: *advantage estimation*. GRPO uses **prompt-level (local) normalization**, estimating a baseline and standard deviation from responses to one prompt; RLOO uses a leave-one-out baseline without inherently requiring this scale normalization. These are distinct design choices. First, Appendix B shows that finite-sample self-normalization differs from population standardization. Second, small groups (e.g., $k = 4$ or $k = 8$) make the scale sensitive to sampled outcomes. Homogeneous groups provide no centered reward signal, while different reward spreads induce different prompt-specific scales. Finally, these relative comparisons

may emphasize fitting individual groups rather than transfer across prompts. Our experiments examine the resulting training dynamics and held-out generalization.

We introduce two distinct variants of this framework, each tailored to a specific use case. REINFORCE++ is an efficient algorithm with $k \geq 1$. In its $k = 1$ configuration (detailed in Algorithm 1), it maximizes prompt diversity for general-domain RLHF. It can also be applied with $k > 1$ (Section 4.2), using the same global normalization principle. REINFORCE++_{w/ Baseline} is a specialized *variant* for complex tasks ($k > 1$) that benefits from **group sampling**. It fixes the flaws of GRPO by first subtracting the *group mean* (for reward reshaping) and then normalizing by the *global standard deviation* (for stability). It also employs a k_2 gradient surrogate for separate KL regularization, as detailed in Appendix C. In summary, our contributions are as follows:

- We provide a theoretical proof that finite-sample local normalization differs from population standardization under explicit sampling assumptions (Appendix B).
- We propose REINFORCE++, a critic-free framework centered on Global Advantage Normalization, as a stable, efficient, and theoretically sound alternative.
- We present two specialized algorithms: We can use REINFORCE++ when $k \geq 1$ for efficient general RLHF and reasoning, and use REINFORCE++_{w/ Baseline} when $k > 1$ for robust complex agentic tasks.
- The experiments show that REINFORCE++ achieves competitive general RLHF scores with shorter responses and stronger held-out reasoning performance. At the same time, REINFORCE++_{w/ Baseline} outperforms both GRPO and PPO in complex agentic reasoning tasks.

2 BACKGROUND AND RELATED WORK

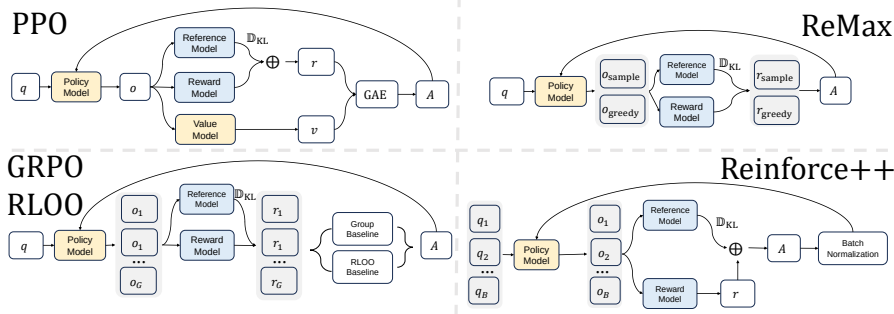


Figure 1: The comparison of PPO, ReMax, GRPO, RLOO, and REINFORCE++. REINFORCE++, as shown in the figure, removes the critic model and uses global batch normalization. REINFORCE++_{w/ Baseline} uses group sampling, similar to GRPO/RLOO, but replaces the local group baseline with a group-mean subtraction followed by global batch normalization.

2.1 PPO AND CRITIC-FREE RLHF

PPO optimizes LLMs by maximizing the following surrogate objective:

$$\mathcal{L}_{PPO}(\theta) = \mathbb{E}_{q \sim P(Q), o \sim \pi_{\theta_{old}}(O|q)} \left[\frac{1}{|o|} \sum_{t=1}^{|o|} \min \left(s_t(\theta) A_t, \text{clip}(s_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t \right) \right] \quad (1)$$

where $s_t(\theta) = \frac{\pi_\theta(o_t|q, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t|q, o_{<t})}$ is the probability ratio. The advantage A_t is typically calculated using Generalized Advantage Estimation (GAE) (Schulman et al., 2015):

$$A_{q,o_t} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l} \quad (2)$$

where $\delta_{q,o_t} = r_t + \gamma V(o_{t+1}) - V(o_t)$ is the temporal difference error, and $V(\cdot)$ is the critic network. Critic-free methods in Figure 1 remove the critic $V(\cdot)$ and compute the advantage A_t directly from rewards.

- **ReMax** (Li et al., 2023) uses a greedy decoding response $\hat{o}$ as the baseline: $A_{q,o_t} = r(o) - r(\hat{o})$.
- **RLOO** (Ahmadian et al., 2024) samples k responses and uses the mean of others as the baseline: $A_{q,o_t^{(i)}} = r(o^{(i)}) - \frac{1}{k-1} \sum_{j \neq i} r(o^{(j)})$.
- **GRPO** (Shao et al., 2024) also samples k responses but normalizes the advantage using the mean and standard deviation of the *local group*:

$$A_{q,o_t^{(i)}} = \frac{r(o^{(i)}) - \text{mean}(\{r(o^{(j)})\}_{j=1}^k)}{\text{std}(\{r(o^{(j)})\}_{j=1}^k) + \epsilon} \quad (3)$$

2.2 THE PROBLEM WITH LOCAL NORMALIZATION

The core issue with methods like GRPO lies in Eq. (3). The usage of **prompt-level (local) normalization** is problematic for three reasons:

1. **Finite-Sample Distortion:** Appendix B proves that self-normalization differs from the population-standardized reward in finite samples. This characterizes the scalar normalization mechanism; it does not by itself establish bias of the full policy-gradient estimator.
2. **Practical Sensitivity:** Small groups (e.g., 4 or 8 responses) provide noisy estimates of reward spread. Standardized advantages are bounded by $\sqrt{k-1}$ for nonconstant groups using divisor k , rather than exploding as the spread vanishes. Identical rewards produce zero centered signal; mixed outcomes induce prompt-specific update scales.
3. **Task Overfitting:** Group-relative updates compare responses to the same prompt. The resulting prompt weighting may affect transfer to other tasks. Our small-data experiments distinguish training-set fit from held-out performance rather than treating overfitting as an inevitable consequence.

3 METHOD

Our method addresses the instability of critic-free RLHF by replacing biased local normalization with stable global normalization. We propose two variants for different use cases.

3.1 GLOBAL NORMALIZATION

The primary REINFORCE++ algorithm is designed for general-purpose RLHF. In its $k = 1$ configuration, it is designed for maximum efficiency and prompt diversity.

It optimizes the PPO objective in Eq. (1) but redefines the advantage A_{q,o_t} . We use the standard PPO reward formulation, which incorporates the kl-style KL penalty directly into the reward:

$$A_{q,o_t} = r(o_{1:T}, q) - \beta \cdot \sum_{i=t}^T \text{KL}(i) \quad (4)$$

where $\text{KL}(t) = \log \left(\frac{\pi_{\theta_{\text{old}}}(o_t|q, o_{<t})}{\pi_{\text{ref}}(o_t|q, o_{<t})} \right)$.

The key innovation is our normalization strategy. Instead of a non-existent (for $k = 1$) or biased local norm, we use Global Advantage Normalization (Andrychowicz et al., 2020):

$$A_{q,o_t}^{\text{norm}} = \frac{A_{q,o_t} - \text{mean}(A|A \in \mathcal{D}_{\text{batch}})}{\text{std}(A|A \in \mathcal{D}_{\text{batch}}) + \epsilon} \quad (5)$$

The global normalization is the core of REINFORCE++. Pooling a large rollout batch (e.g., 1024 responses) replaces separate small-group scales with shared statistics. Under independent sampling, sample moments converge to population values (Appendix B); this is not a general unbiased-policy-gradient guarantee. Valid token positions exclude padding, and the moments remain fixed during rollout reuse. The efficient $k = 1$ case is detailed in Algorithm 1.

3.2 LOCAL BASELINE

For more complex tasks, such as multi-step reasoning, sampling multiple responses per prompt ($k > 1$) can be beneficial (Yue et al., 2025). To this end, we introduce REINFORCE++_{w/ Baseline}. This variant combines the benefits of group-sampling with the stability of global normalization. The advantage calculation is a two-step process:

- **Group Mean Subtraction for Reshaping:** We first subtract the group mean reward. It serves as a local baseline, reshaping rewards to be robust to different reward scales (e.g., 0/1 vs. -1/1).

$$A'_{q,o_t^{(i)}} = r(q, o_t^{(i)}) - \frac{1}{k} \sum_{j=1}^k r(q, o_t^{(j)}) \quad (6)$$

- **Global Batch Normalization for Stability:** We then normalize this initial advantage using the **global batch statistics**, not the unstable local group statistics.

$$A_{q,o_t}^{\text{norm}} = \frac{A'_{q,o_t} - \text{mean}_{\text{batch}}(A')}{\text{std}_{\text{batch}}(A') + \epsilon} \quad (7)$$

This separates reward centering from scale estimation; the inclusive group baseline equals $(k - 1)/k$ times RLOO (Appendix D).

Furthermore, this variant employs a separate k_2 KL regularizer. At a fixed context with on-policy actions held fixed during differentiation, its expected pointwise gradient equals the reverse-KL gradient (Liu et al., 2025a). This equivalence concerns a gradient surrogate, not KL values; unweighted old-policy rollout reuse is off-policy (Appendix C). The maximized objective is:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}}(A^{\text{norm}}) - \lambda \mathbb{E}_{o \sim \pi_{\text{old}}} \left[\frac{1}{|o|} \sum_t \frac{1}{2} \left(\log \frac{\pi_{\theta}(o_t | q, o_{<t})}{\pi_{\text{ref}}(o_t | q, o_{<t})} \right)^2 \right]. \quad (8)$$

3.3 RELATIONSHIP WITH PPO

Both variants retain PPO clipping without a learned critic. Setting $V = 0$ and $\gamma = \lambda_{\text{GAE}} = 1$ reduces GAE to Monte Carlo returns, the structure used by the basic variant before normalization. REINFORCE++_{w/ Baseline} additionally uses prompt-specific reward centering and a separate KL loss; these changes distinguish it from standard actor-critic PPO.

3.4 IMPLEMENTATION TRICKS

Token-level Advantage The task’s immediate reward is zero before termination, but its undiscounted return contributes to every preceding response token in Eq. (4). REINFORCE++_{w/ Baseline} broadcasts its centered terminal reward to valid response positions. The stated objective averages token losses within each response and then across responses, as specified in Appendix A.

Batch Construction For REINFORCE++ ($k = 1$), a global batch of size N comprises N distinct prompts. For REINFORCE++_{w/ Baseline} ($k = 4$), a global batch of size $N = 1024$ might contain $1024/4 = 256$ unique prompts. The global moments pool valid response positions across the rollout batch.

Mini-Batch Updates To enhance training efficiency, we implement mini-batch updates with the following characteristics:

- **Batch Processing:** Data is processed in smaller, manageable chunks rather than full-batch updates.
- **Multiple Updates:** Reuse detached advantages and old-policy probabilities over E optimization epochs.
- **Stochastic Optimization:** Introduces beneficial randomness for better generalization.

Reward Normalization and Clipping We implement comprehensive reward processing to stabilize training:

- **Normalization:** Uses a shared z-score scale across the batch; standardization alone does not remove outliers.
- **Clipping:** Constrains reward values within predefined bounds to avoid instability.
- **Scaling:** Applies appropriate scaling factors for numerical stability during updates.

3.5 WHY USE GLOBAL BATCH NORMALIZATION?

Omitting the numerical stabilizer, under independent sampling with nonzero population variance, normalization approaches the population-standardized target as $N \rightarrow \infty$ (Appendix B). This motivates global normalization in both REINFORCE++ and REINFORCE++_{w/ Baseline}: a batch of 1024 responses pools more observations than groups of 4 or 8. Token dependence and repeated responses limit the effective independent sample count. The result explains the scale estimate, rather than proving that every clipped, reused-rollout policy gradient is unbiased.

3.6 SUMMARY

To address these challenges, we propose REINFORCE++, a method centered on a simple yet powerful idea: **Global Advantage Normalization**, without a critic. Instead of normalizing within a prompt’s local group, REINFORCE++ normalizes the advantage function across the *entire global training batch*. The overall approach is:

- **Theoretical Stability:** With independent sampling, increasing the batch size improves population-moment estimation. Finite-sample effects and trajectory dependence remain relevant (Appendix B).
- **Training Efficiency:** It retains the critic-free architecture, significantly reducing computational and memory overhead compared to PPO.
- **Strong Generalization:** Using a shared scale is associated with improved held-out reasoning results in our evaluated settings.

Algorithm 1 REINFORCE++ when $k = 1$

Require: Initial policy model π_{ref} , reward models R , task prompts $\mathcal{D}$

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1: policy model  $\pi_\theta \leftarrow \pi_{ref}$ 
2: for step = 1, ...,  $M$  do
3:   Sample a batch  $\mathcal{D}_{batch}$  from  $\mathcal{D}$ 
4:   Update the old policy model  $\pi_{old} \leftarrow \pi_\theta$ 
5:   Sample output ( $k = 1$ )  $o \sim \pi_{old}(\cdot | q)$  for each question  $q \in \mathcal{D}_{batch}$ 
6:   Compute rewards  $r_i$  for each sampled  $o_i$  using  $R$ 
7:   Compute advantage  $A_{q,o_t}$  via Eq. (4)
8:   Normalize advantages globally across  $\mathcal{D}_{batch}$  to get  $A_{q,o_t}^{norm}$  via Eq. (5)
9:   for iteration = 1, ...,  $E$  do
10:    Update  $\pi_\theta$  by maximizing the objective in Eq. (1) using  $A_{q,o_t}^{norm}$ 
11:   end for
12: end for
Ensure:  $\pi_\theta$ 

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Table 1: Comparison on Chat-Arena-Hard (Li et al., 2024). The single-sample REINFORCE++ ($k = 1$) achieves a top-tier score with a higher score/length ratio than group-sampling GRPO.

	Score	Length	Per Token
REINFORCE++ ($k=1$)	46.7	832	0.0561
GRPO ($k=4$)	46.8	860	0.0544
RLOO ($k=4$)	44.6	866	0.0515
ReMax ($k=1+1$)	45.1	805	0.0560

4 EXPERIMENTS

We evaluate our two algorithms, REINFORCE++ and REINFORCE++_{w/ Baseline}, in their respective target domains using the OpenRLHF framework (Hu et al., 2024).

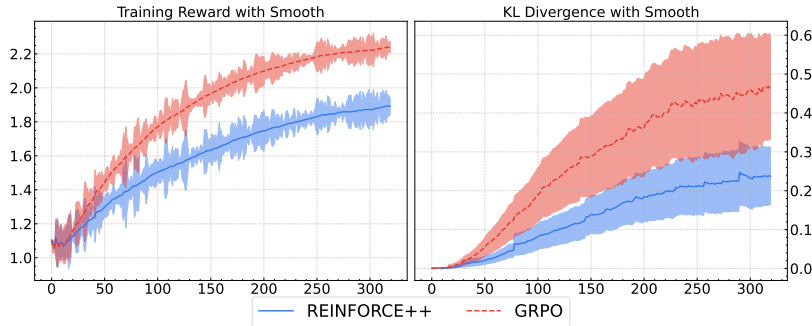


Figure 2: Comparison of smoothed Training Reward and KL Divergence. REINFORCE++ ($k = 1$) achieves strong rewards with significantly more stable (lower) KL divergence, illustrating a different reward-KL trade-off.

4.1 GENERAL RLHF

First, we evaluate the REINFORCE++ in its $k = 1$ configuration, focusing on general-domain RLHF, where efficiency and prompt diversity are key.

Experimental Setup We use an instruction-following policy model (Llama-3-8B-SFT) and refine it using a Bradley-Terry reward model (Bai et al., 2022) trained on $\sim 700K$ human preference pairs. The policy is trained on 20,000 diverse prompts. We compare REINFORCE++ ($k = 1$) with other critic-free methods: GRPO ($k = 4$), RLOO ($k = 4$), and ReMax ($k = 1 + 1$).

Experimental Results As shown in Table 1, the single-sample REINFORCE++ ($k = 1$) achieves a score of 46.7, close to the group-sampling GRPO (46.8). However, it produces shorter responses (832 tokens vs. 860), yielding a higher score/length ratio (0.0561). This result supports competitive alignment with single-response sampling; score/length is not a direct measure of training compute efficiency.

Results Analysis Figure 2 shows the training dynamics. GRPO’s reward and KL divergence rise rapidly, while REINFORCE++ exhibits a more gradual reward increase and smaller divergence. These curves describe different reward-KL trade-offs; they do not alone establish reward hacking. Response length can also influence preference-based evaluation (Dubois et al., 2024). We therefore report both evaluation scores and generated lengths rather than equating higher training reward with better generalization.

Table 2: Comparison on a small training dataset. GRPO (local norm) shows a large train–test gap, while REINFORCE++ (global norm) generalizes.

Pass@N	AIME-24 (Train)	AIME-25 (Test)	
	N = 1	N = 1	N = 16
GRPO	95.0	0.0	0.4
REINFORCE++	71.0	2.5	40.0

Table 3: Comparison on RL from Zero. REINFORCE++ shows superior OOD performance.

Pass@N	AIME-24 (OOD)	AMC-23 (OOD)	MATH-500 (ID)
	N = 8	N = 8	N = 1
GRPO	18.96	59.22	73.00
REINFORCE++	21.04	60.47	72.00

4.2 REASONING EXPERIMENTS

Next, we evaluate REINFORCE++ (group-sample, $k > 1$) in its target domain: complex reasoning, where group sampling is standard, and reward signals are often sparse (e.g., 0/1). We compare it directly with GRPO ($k > 1$).

4.2.1 LONG REASONING TASK

We use rule-based rewards (RLVR) in mathematical reasoning settings (Guo et al., 2025; Seed et al., 2025).

Analysis on Small-Scale Datasets To test for overfitting, we trained on only 30 questions from AIME-24 and evaluated on AIME-25. As shown in Table 2, GRPO (local norm) achieves a near-perfect 95.0% on the training set but scores only 0.0 Pass@1 and 0.4 Pass@16 on the test set, indicating severe overfitting. In contrast, REINFORCE++ with global norm, while scoring lower on the training set (71.0), shows higher held-out scores (2.5 Pass@1, 40.0 Pass@16).

The curves show the same overfitting pattern: GRPO fits the training questions quickly, while REINFORCE++ learns more gradually.

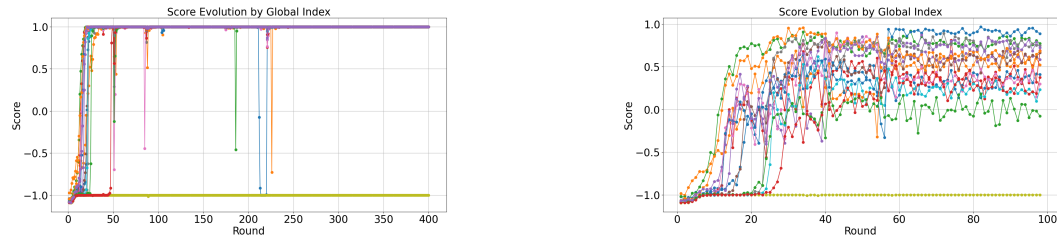


Figure 3: Training curves on a small prompt dataset. Left: GRPO (local norm) immediately overfits. Right: REINFORCE++ (global norm) learns more stably.

Logical Reasoning (K&K Puzzles) We also tested on the Knights and Knaves (K&K) puzzles (Xie et al., 2025; 2024), where difficulty increases with the number of "people." Figure 4 shows that while GRPO is competitive on easy tasks (2-3 people), its performance collapses on harder, OOD tasks (8 people). REINFORCE++ is more robust, outperforming GRPO on all tasks with four or more people and achieving a much higher average score (62.1 vs. 55.7).

RL from Zero Setting Finally, we trained a Qwen2.5-Math-Base model from zero on MATH dataset splits (Guo et al., 2025). Table 3 shows that REINFORCE++ again achieves better OOD generalization on the more challenging AIME-24 and AMC-23 datasets, while remaining competitive on the in-distribution MATH-500 test set.

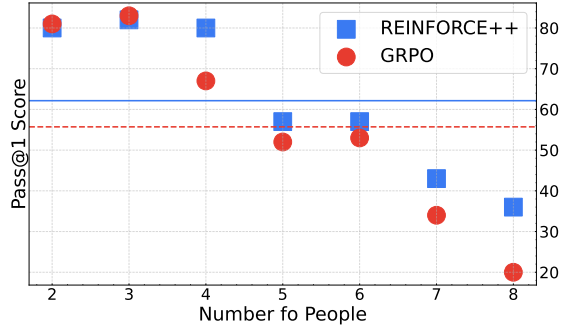


Figure 4: Comparison on logic benchmarks. REINFORCE++’s global normalization provides better robustness to increasing task difficulty and OOD challenges (8 people).

Table 4: Performance Comparison on complex tool-use benchmarks (average@32). REINFORCE++_{w/ Baseline} outperforms both GRPO (local norm) and PPO (critic-based) models.

Algorithm	AIME 24	AIME 25	HMMT 2025	HMMT 2024	CMIMC	Avg
GRPO (local norm)	31.66	21.87	16.97	17.70	24.68	22.58
PPO (critic-based)	30.20	21.66	15.00	18.43	23.95	21.85
RF++-Baseline (global norm)	30.83	27.18	17.91	18.95	25.62	24.10

4.3 MULTI-STEP REINFORCEMENT LEARNING

Finally, to test performance in a complex, multi-step environment, we use the zero-shot agent setup from Mai et al. (2025), in which a Qwen 2.5 Base 7B model must learn to use Python tools to solve mathematical problems. It is a challenging scenario where group sampling ($k > 1$) and reward reshaping are crucial (see Section 5).

Experimental Setup We adhere to the training and evaluation protocols of the ZeroTIR environment Mai et al. (2025). The backbone model is Qwen 2.5 Base 7B, trained using OpenRLHF Hu et al. (2024) with datasets from ORZ Hu et al. (2025) and DAPO Yu et al. (2025). We assess performance on benchmarks including AIME 2024 Jia (2024), AIME-2025 math ai (2025), HMMT FEB-2024/2025, and CMIMC, using the average@32 metric.

Experimental Result As shown in Table 4, REINFORCE++_{w/ Baseline} achieves the highest average accuracy (24.10) across all benchmarks. It outperforms both GRPO (22.58) and the full-critic PPO (21.85). It demonstrates that our stable, critic-free approach is highly effective for complex agentic tasks, surpassing even the heavyweight PPO algorithm. The combined recipe leads on four of five tasks; this comparison does not isolate the individual effects of reward centering, normalization, and KL regularization.

5 BEST PRACTICES

5.1 GENERAL PRINCIPLE

This paper introduces two algorithms: REINFORCE++ and REINFORCE++_{w/ Baseline}, with a key focus on selecting the appropriate algorithm based on task conditions.

Empirical evidence from the open-source community suggests that REINFORCE++_{w/ Baseline} is particularly effective for sample filtering or more complex scenarios. For instance, in the multi-turn tool-calling setting, a high proportion of void (i.e., non-informative or incorrect) samples can destabilize training. In such cases, incorporating a baseline by subtracting the intra-group mean reward

significantly improves training stability by effectively filtering out void samples. Moreover, this automatic reward reshaping reduces the need for designing complex reward structures. For example, REINFORCE++_{w/ Baseline} supports both 0/1 and -1/1 reward schemes, while REINFORCE++ performs best with symmetric rewards, such as -1/1 in RLVR tasks.

In contrast, for general-domain tasks where prompt diversity and efficiency are paramount, or for tasks where obtaining multiple reward signals for distinct responses is challenging—such as training with Process-Supervised Reward Models (PRMs) or online real-time sampling—we recommend using REINFORCE++ ($k = 1$). Our experiments (Section 4.1) show competitive preference alignment with shorter responses in this setting.

5.2 THIRD-PARTY VALIDATION

The core principle of REINFORCE++, global advantage normalization, has been independently validated and adopted in several **subsequent** large-scale reasoning systems, confirming its stability and effectiveness.

LitePPO (Liu et al., 2025c), which effectively combines REINFORCE++_{w/ Baseline} with a token-level loss, conducted experiments demonstrating that the global standard deviation is superior to the local standard deviation used in GRPO. Their results indicate more stable training and better generalization, which aligns with our findings.

ScaleRL (Khatri et al., 2025) performed a detailed ablation study on advantage estimation methods in large-scale (16,000 GPU-hour) experiments. They directly compared batch-level normalization in REINFORCE++ with prompt-level normalization in GRPO. Their findings concluded that batch-level normalization was “slightly superior in both compute efficiency and final performance,” confirming the advantages of global normalization at scale.

DLER (Liu et al., 2025b) found that when using truncation to control the output length of an LLM, batch-wise (global) normalization remains stable while group-wise (local) normalization shows declining accuracy. The study provides further evidence that global normalization is more robust to variations in training conditions and reward landscapes.

6 CONCLUSION

In this paper, we present REINFORCE++, a critic-free RLHF framework designed to enhance training stability by addressing fundamental flaws in existing methods. We identified that prior critic-free algorithms, such as GRPO, rely on *prompt-level (local)* advantage normalization, whose finite-sample distortion we characterize (Appendix B) and demonstrate to be practically unstable.

Our solution, **Global Advantage Normalization**, normalizes advantages across the entire batch, providing a shared scale that converges to population standardization under independent sampling. We proposed two variants tailored for different needs:

1. REINFORCE++ : For general-domain RLHF, we show the single-sample ($k = 1$) approach achieves competitive preference alignment with broader prompt coverage. We also show that its application with $k > 1$ is robust for reasoning tasks (Section 4.2).
2. REINFORCE++_{w/ Baseline} : For complex reasoning and agent tasks, this variant combines group-mean reshaping with stable global normalization and a k_2 gradient surrogate for separate KL regularization.

Our empirical results, supported by independent validation from multiple third-party systems, confirmed the effectiveness of this specialized approach. REINFORCE++ demonstrated competitive general RLHF results and stronger held-out reasoning scores. The basic variant improved held-out performance in the small-data setting, while REINFORCE++_{w/ Baseline} outperformed both GRPO and full-critic PPO in complex, long-horizon tool-use tasks. This work demonstrates that a theoretically sound, stable approach without a critic model can be more efficient and effective than traditional PPO.

AI USE STATEMENT

Generative AI tools assisted with template migration, formatting, compilation diagnostics, and mathematical exposition and proof revision. They were not used to generate experimental data, fabricate results, or conduct new experiments. The authors reviewed all AI-assisted work and take responsibility for the final content of this paper, including its text, claims, and artifacts.

ETHICS STATEMENT

Reinforcement learning is increasingly used to steer large language models by optimizing them against a reward signal, including verifiable, programmatic rewards (RLVR) for domains such as math, code, and tool use. The main potential positive impact of this line of work is improved scalability and reproducibility of post-training: verifiable rewards can reduce reliance on human labeling and make optimization targets easier to audit and replicate. By improving the stability and efficiency of critic-free optimization under sparse and brittle rewards, our method can help make RLVR-style training more reliable.

Potential risks include reward gaming when verifiers are incomplete or mis-specified, and dual-use concerns if improved tool-use or problem-solving capabilities are deployed without safeguards. We recommend evaluating beyond verifier scores (e.g., robustness and red-teaming), stress-testing verifiers for exploitability, and using conservative deployment controls (e.g., access restrictions and monitoring) for systems trained with RL.

REPRODUCIBILITY STATEMENT

Section 3 specifies the algorithms, and Section 4 describes the experimental settings. Appendix A specifies the response-wise loss reduction of the stated objective. Appendices B and C provide assumptions and derivations for normalization and KL regularization.

REFERENCES

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Arash Ahmadian, Chris Cremer, Matthias Gallé, Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet stün, and Sara Hooker. Back to basics: Revisiting reinforce style optimization for learning from human feedback in llms. *arXiv preprint*, 2024.
- Marcin Andrychowicz, Anton Raichuk, Piotr Stańczyk, Manu Orsini, Sertan Girgin, Raphael Marinier, Léonard Hussenot, Matthieu Geist, Olivier Pietquin, Marcin Michalski, et al. What matters in on-policy reinforcement learning? a large-scale empirical study. *arXiv preprint arXiv:2006.05990*, 2020.
- AI Anthropic. Introducing claude, 2023. URL <https://www.anthropic.com/news/introducing-claude>.
- Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.
- Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B Hashimoto. Length-controlled alpacaEval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*, 2024.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Jian Hu, Xibin Wu, Zilin Zhu, Weixun Wang, Dehao Zhang, Yu Cao, et al. Openrlhf: An easy-to-use, scalable and high-performance rlhf framework. *arXiv preprint arXiv:2405.11143*, 2024.

- Jingcheng Hu, Yinmin Zhang, Qi Han, Daxin Jiang, Xiangyu Zhang, and Heung-Yeung Shum. Open-reasoner-zero: An open source approach to scaling up reinforcement learning on the base model. *arXiv preprint arXiv:2503.24290*, 2025.
- Maxwell Jia. Aime 2024 dataset card. https://huggingface.co/datasets/Maxwell-Jia/AIME_2024, 2024. Version 2024-03.
- Devvrit Khatri, Lovish Madaan, Rishabh Tiwari, Rachit Bansal, Sai Surya Duvvuri, Manzil Zaheer, Inderjit S Dhillon, David Brandfonbrener, and Rishabh Agarwal. The art of scaling reinforcement learning compute for llms. *arXiv preprint arXiv:2510.13786*, 2025.
- Tianle Li, Wei-Lin Chiang, Evan Frick, Lisa Dunlap, Tianhao Wu, Banghua Zhu, Joseph E Gonzalez, and Ion Stoica. From crowdsourced data to high-quality benchmarks: Arena-hard and benchbuilder pipeline. *arXiv preprint arXiv:2406.11939*, 2024.
- Ziniu Li, Tian Xu, Yushun Zhang, Yang Yu, Ruoyu Sun, and Zhi-Quan Luo. Remax: A simple, effective, and efficient method for aligning large language models. *arXiv preprint arXiv:2310.10505*, 2023.
- Kezhao Liu, Jason Klein Liu, Mingtao Chen, and Yiming Liu. Rethinking kl regularization in rlhf: From value estimation to gradient optimization. *arXiv preprint arXiv:2510.01555*, 2025a.
- Shih-Yang Liu, Xin Dong, Ximing Lu, Shizhe Diao, Mingjie Liu, Min-Hung Chen, Hongxu Yin, Yu-Chiang Frank Wang, Kwang-Ting Cheng, Yejin Choi, et al. Dler: Doing length penalty right-incentivizing more intelligence per token via reinforcement learning. *arXiv preprint arXiv:2510.15110*, 2025b.
- Zihe Liu, Jiashun Liu, Yancheng He, Weixun Wang, Jiaheng Liu, Ling Pan, Xinyu Hu, Shaopan Xiong, Ju Huang, Jian Hu, et al. Part i: Tricks or traps? a deep dive into rl for llm reasoning. *arXiv preprint arXiv:2508.08221*, 2025c.
- Xinji Mai, Haotian Xu, Zhong-Zhi Li, Weinong Wang, Jian Hu, Yingying Zhang, Wenqiang Zhang, et al. Agent rl scaling law: Agent rl with spontaneous code execution for mathematical problem solving. *arXiv preprint arXiv:2505.07773*, 2025.
- math ai. Aime 2025 dataset card. <https://huggingface.co/datasets/math-ai/aime25>, 2025. Version 2024-03.
- OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *arXiv preprint arXiv:2203.02155*, 2022.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2305.18290>.
- John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. *arXiv preprint arXiv:1506.02438*, 2015.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- ByteDance Seed, Jiaze Chen, Tiantian Fan, Xin Liu, Lingjun Liu, Zhiqi Lin, Mingxuan Wang, Chengyi Wang, Xiangpeng Wei, Wenyuan Xu, et al. Seed1. 5-thinking: Advancing superb reasoning models with reinforcement learning. *arXiv preprint arXiv:2504.13914*, 2025.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, YK Li, Yu Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

- Wei Shen and Chuheng Zhang. Policy filtration in rlhf to fine-tune llm for code generation. *arXiv preprint arXiv:2409.06957*, 2024.
- Wei Shen, Guanlin Liu, Zheng Wu, Ruofei Zhu, Qingping Yang, Chao Xin, Yu Yue, and Lin Yan. Exploring data scaling trends and effects in reinforcement learning from human feedback. *arXiv preprint arXiv:2503.22230*, 2025.
- Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- Sai Vemprala, Rogerio Bonatti, Arthur Buckner, and Ashish Kapoor. Chatgpt for robotics: Design principles and model abilities. *Microsoft Auton. Syst. Robot. Res.*, 2:20, 2023.
- Chulin Xie, Yangsibo Huang, Chiyuan Zhang, Da Yu, Xinyun Chen, Bill Yuchen Lin, Bo Li, Badih Ghazi, and Ravi Kumar. On memorization of large language models in logical reasoning. *arXiv preprint arXiv:2410.23123*, 2024.
- Tian Xie, Zitian Gao, Qingnan Ren, Haoming Luo, Yuqian Hong, Bryan Dai, Joey Zhou, Kai Qiu, Zhirong Wu, and Chong Luo. Logic-rl: Unleashing llm reasoning with rule-based reinforcement learning. *arXiv preprint arXiv:2502.14768*, 2025.
- Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- Yu Yue, Yufeng Yuan, Qiyang Yu, Xiaochen Zuo, Ruofei Zhu, Wenyuan Xu, Jiase Chen, Chengyi Wang, TianTian Fan, Zhengyin Du, et al. Vapo: Efficient and reliable reinforcement learning for advanced reasoning tasks. *arXiv preprint arXiv:2504.05118*, 2025.

A RESPONSE-WISE LOSS REDUCTION AND IMPLEMENTATION

The objective in Eq. (1) uses a mean over valid tokens within each response, followed by a mean over responses in the batch. This section specifies the reduction independently of advantage normalization.

Let $m_{i,t} \in \{0, 1\}$ indicate a valid response token included in the policy loss, and let $n_i = \sum_t m_{i,t} > 0$. For B responses, the policy loss is

$$L_{\text{PG}}(\theta) = \frac{1}{B} \sum_{i=1}^B \left[\frac{1}{n_i} \sum_t m_{i,t} \ell_{i,t}(\theta) \right], \quad \ell_{i,t}(\theta) = -f_{i,t}(\theta, \text{sg}[\hat{A}_{i,t}]). \quad (9)$$

Here $\hat{A}$ denotes A^{norm} , f is the clipped surrogate in Eq. (1), and sg denotes stop-gradient. Padding and positions outside the response loss mask do not enter the numerator or token count. The separate k_2 term in Eq. (8) uses the same reduction.

Response weighting. Each response has outer weight $1/B$, and each valid token within response i has weight $1/(Bn_i)$. This differs from dividing the sum of all token losses by the total number of valid tokens:

$$L_{\text{token}} = \frac{\sum_{i,t} m_{i,t} \ell_{i,t}}{\sum_i n_i}. \quad (10)$$

The latter weights each response in proportion to its length. The two reductions agree when all responses have equal length. With a fixed number k of responses per prompt, batch averaging is also equivalent to averaging the per-response losses within each prompt and then across prompts.

Minibatches and accumulation. For a minibatch $\mathcal{M}$, replace B by $|\mathcal{M}|$ in Eq. (9). To recover the same mean when combining uneven microbatches or distributed shards, their loss sums must be weighted by their response counts. The normalization moments are associated with the rollout batch; changing the partition into minibatches does not change the definition of the already computed advantages.

Terminal reward and token returns. For the basic variant, write $d_{i,t} = \log(\pi_{\text{old}}/\pi_{\text{ref}})$ at the sampled token. The immediate reward can be written as

$$\tilde{r}_{i,t} = \mathbf{1}\{t = T_i\}r(q_i, o_i) - \beta d_{i,t}. \quad (11)$$

With unit discount, its return is $\sum_{u=t}^{T_i} \tilde{r}_{i,u} = A_{q_i, o_{i,t}}$ from Eq. (4). Thus, a terminal task reward contributes to the return at every preceding response position. For REINFORCE++_{w/ Baseline}, the group-centered terminal reward in Eq. (6) is broadcast to response positions before global normalization.

B FINITE-SAMPLE PROPERTIES OF NORMALIZATION

We analyze a scalar model to make the effect of estimating normalization statistics from the same samples explicit. Let

$$r_i = \mu + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma > 0, \quad (12)$$

and define $\bar{\epsilon} = N^{-1} \sum_i \epsilon_i$, $D_N^2 = N^{-1} \sum_i (\epsilon_i - \bar{\epsilon})^2$, and $Z_{i,N} = (\epsilon_i - \bar{\epsilon})/D_N$ for $N \geq 2$. The population-standardized target is ϵ_i/σ . We omit the numerical stabilizer for this analysis; $D_N > 0$ almost surely under the stated model.

Proposition B.1 (Finite-sample bound). *For any nonconstant sample of size $N \geq 2$, $|Z_{i,N}| \leq \sqrt{N-1}$. Consequently, under the Gaussian model, $\mathbb{E}[Z_{i,N} \mid \epsilon_i]$ cannot equal ϵ_i/σ almost surely for finite N .*

Proof. Set $x_j = \epsilon_j - \bar{\epsilon}$, so $\sum_j x_j = 0$. By Cauchy–Schwarz,

$$x_i^2 = \left(\sum_{j \neq i} x_j \right)^2 \leq (N-1) \sum_{j \neq i} x_j^2. \quad (13)$$

Therefore $\sum_j x_j^2 \geq Nx_i^2/(N-1)$, giving $Z_{i,N}^2 = Nx_i^2/\sum_j x_j^2 \leq N-1$. The conditional expectation obeys the same bound, whereas $|\epsilon_i|/\sigma > \sqrt{N-1}$ has positive probability for a Gaussian variable. Equality with the standardized target cannot hold almost surely. $\square$

Conditional moments of the numerator and denominator. The centered numerator satisfies

$$\mathbb{E}[\epsilon_i - \bar{\epsilon} \mid \epsilon_i] = \left(1 - \frac{1}{N}\right) \epsilon_i. \quad (14)$$

Using $D_N^2 = N^{-1} \sum_j \epsilon_j^2 - \bar{\epsilon}^2$ and independence,

$$\mathbb{E}[\bar{\epsilon}^2 \mid \epsilon_i] = \frac{\epsilon_i^2 + (N-1)\sigma^2}{N^2}, \quad (15)$$

$$\mathbb{E}[D_N^2 \mid \epsilon_i] = \frac{(N-1)^2}{N^2} \sigma^2 + \frac{N-1}{N^2} \epsilon_i^2. \quad (16)$$

These identities show the conditional dependence of the normalization statistics on the included observation. They do not permit replacing the expectation of a ratio by a ratio of expectations; Proposition B.1 establishes the finite-sample discrepancy directly.

An exact two-response example. For $N=2$, $D_2 = |\epsilon_1 - \epsilon_2|/2$ and $Z_{1,2} = \text{sign}(\epsilon_1 - \epsilon_2)$ almost surely. Thus,

$$\mathbb{E}[Z_{1,2} \mid \epsilon_1 = x] = 2\Phi(x/\sigma) - 1, \quad (17)$$

where Φ is the standard normal cumulative distribution function. This bounded, nonlinear function differs from the linear population-standardized value x/σ . It illustrates how a small group changes the reward scale through self-normalization.

Proposition B.2 (Large-sample limit). *Under the Gaussian model, for any fixed i , $Z_{i,N} \rightarrow \epsilon_i/\sigma$ almost surely and in L^1 as $N \rightarrow \infty$. In particular,*

$$\mathbb{E}[|\mathbb{E}[Z_{i,N} \mid \epsilon_i] - \epsilon_i/\sigma|] \rightarrow 0. \quad (18)$$

Proof. The strong law gives $\bar{\epsilon} \rightarrow 0$ and $D_N^2 \rightarrow \sigma^2$ almost surely, establishing the first limit. By exchangeability and $\sum_{j=1}^N Z_{j,N}^2 = N$, we have $\mathbb{E}[Z_{i,N}^2] = 1$ for every N . This uniform second-moment bound implies uniform integrability, so the almost-sure convergence also holds in L^1 . Conditional Jensen's inequality gives

$$\mathbb{E}[|\mathbb{E}[Z_{i,N} - \epsilon_i/\sigma \mid \epsilon_i]|] \leq \mathbb{E}[|Z_{i,N} - \epsilon_i/\sigma|] \rightarrow 0. \quad (19)$$

$\square$

Numerical stabilizer. With a fixed stabilizer $\eta > 0$, $(\epsilon_i - \bar{\epsilon})/(D_N + \eta)$ converges instead to $\epsilon_i/(\sigma + \eta)$. The unstabilized population-standardization limit applies when the stabilizer is omitted or tends to zero. The finite-sample bound remains valid with a nonnegative stabilizer for nonconstant samples; constant samples have zero numerator and return zero when the stabilizer is positive.

Implications for global normalization. The propositions isolate the effect of normalization sample size: estimated moments approach their population values as the number of independent observations grows. A shared batch scale can use observations across many prompt groups, rather than estimating a separate scale from each small group. For grouped or token-level data, the interpretation depends on the number of independently sampled trajectories and their dependence structure.

Relationship to baselines and gradients. For an unscaled group mean, $\mathbb{E}[r_i - \bar{r}_q \mid r_i, q] = (1 - 1/k)(r_i - \mu_q)$ when responses are conditionally independent with mean reward μ_q . This is the same factor appearing in Eq. (24). The propositions concern convergence to a standardized scalar target, distinct from unbiasedness of a policy-gradient estimator. Their role is to explain the normalization mechanism under explicit sampling assumptions.

C KL REGULARIZATION AS A GRADIENT SURROGATE

For a fixed context, write $p_\theta(y) = \pi_\theta(y \mid h)$, $q(y) = \pi_{\text{ref}}(y \mid h)$, and $z(y) = \log[p_\theta(y)/q(y)]$. Assume common support and sufficient regularity to interchange differentiation and expectation. The reverse KL is

$$D_{\text{KL}}(p_\theta \| q) = \mathbb{E}_{y \sim p_\theta}[z(y)], \quad \nabla_\theta D_{\text{KL}}(p_\theta \| q) = \mathbb{E}_{p_\theta}[z(y) \nabla_\theta \log p_\theta(y)], \quad (20)$$

using $\mathbb{E}_{p_\theta}[\nabla_\theta \log p_\theta] = 0$.

Value estimators and differentiated losses. The common expressions are

$$k_1(z) = z, \quad k_2(z) = \frac{1}{2}z^2, \quad k_3(z) = e^{-z} - 1 + z. \quad (21)$$

Both k_1 and k_3 have expectation $D_{\text{KL}}(p_\theta \| q)$ under on-policy sampling, since $\mathbb{E}_{p_\theta}[e^{-z}] = 1$. When sampled actions are treated as fixed during automatic differentiation, their pointwise gradients differ:

$$\begin{aligned} \nabla_\theta k_1 &= \nabla_\theta \log p_\theta, \\ \nabla_\theta k_2 &= z \nabla_\theta \log p_\theta, \\ \nabla_\theta k_3 &= (1 - q/p_\theta) \nabla_\theta \log p_\theta. \end{aligned} \quad (22)$$

The on-policy expectation of the k_2 pointwise gradient is exactly Eq. (20). This is the gradient-surrogate interpretation used for the separate KL term in REINFORCE++_{w/ Baseline} (Liu et al., 2025a). It is distinct from differentiating the entire expectation $\mathbb{E}_{p_\theta}[k_2]$, which also differentiates the sampling distribution.

Under the same fixed-context assumptions, the expected pointwise gradient of k_1 is zero. The k_1 expression can instead enter the reward and act as a detached score-function coefficient, as in the returns used by the basic variant. The expected pointwise gradient of k_3 equals $-\mathbb{E}_q[\nabla_\theta \log p_\theta]$, the forward-KL gradient. Its value-estimation and loss-gradient roles should therefore be distinguished.

Numerical sensitivity. The k_3 pointwise gradient contains the importance ratio q/p_θ , which can become large when the current policy assigns low probability to actions favored by the reference. Computing this ratio as $\exp(-z)$ can also overflow for sufficiently negative z . These are potential variance and numerical concerns, not a general proof of infinite variance or a requirement to reset the reference policy. The k_2 surrogate avoids this exponential ratio, but its gradient still depends on the log ratio and score function; the identity above alone does not guarantee low variance or training stability.

Rollout reuse. The on-policy identities apply when the sampled actions follow the current policy. For actions sampled from p_{old} , a fixed-context importance-weighted gradient satisfies

$$\mathbb{E}_{p_{\text{old}}} \left[\frac{p_\theta(y)}{p_{\text{old}}(y)} z(y) \nabla_\theta \log p_\theta(y) \right] = \nabla_\theta D_{\text{KL}}(p_\theta \| q). \quad (23)$$

The practical separate loss in Eq. (8) is the unweighted k_2 surrogate on the rollout data. Its exact gradient-equivalence interpretation applies at the on-policy point; subsequent optimization uses it as a surrogate. For autoregressive policies, the fixed-context statement concerns the conditional next-token distribution.

D DESIGN IMPLICATIONS AND SAMPLING BUDGETS

Local baseline and global scale. For a prompt with $k > 1$ responses, let $C_{q,i} = r_i - k^{-1} \sum_j r_j$. Then

$$C_{q,i} = \frac{k-1}{k} \left(r_i - \frac{1}{k-1} \sum_{j \neq i} r_j \right) = \frac{k-1}{k} A_i^{\text{RLOO}}. \quad (24)$$

The group baseline compares a response with alternatives to the same prompt; global scaling supplies a shared scale across prompts. Homogeneous groups have zero centered task-reward signal. This separates reward centering from prompt-specific standard-deviation normalization.

Response budget. At fixed response count B , single-response sampling covers up to B prompts; grouped sampling covers B/k . Larger groups trade prompt coverage for exploration of each prompt. This does not establish a runtime or memory ratio: response length, batching, optimizer state, and rollout implementation also matter.

Evaluation and attribution. The tool-use metric average@32 averages success across 32 sampled attempts; it differs from pass@32, which counts success if at least one attempt succeeds. The reported tool-use mean is an unweighted average across five benchmarks. The improvements of 1.52 and 2.25 percentage points over GRPO and PPO concern the combined training recipe. Controlled ablations would be needed to separate contributions from normalization, reward centering, KL regularization, and loss reduction.